

Seongwon Jang

Robotics Perception | Computer Vision & ROS2

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TECHNICAL SKILLS

Robotics: ROS2, Gazebo, UR5e, LeRobot, solvePnP, multiview triangulation, Camera-LiDAR calibration

Computer Vision: YOLOv8, ONNX, VideoMAE, cross attention, ReID, person tracking

LLM Systems: SFT, vLLM, MCP, DeepEval Synthesizer, LLM-as-a-judge

EXPERIENCE

Research Intern

Jul 2025 - Sep 2025

Ulsan National Institute of Science and Technology (UNIST)

South Korea

- Reviewed prior algorithms, proposed and implemented a new approach, and compared results using consistent benchmarks.

SELECTED PROJECTS

AI for Industry Challenge | Team Lead | Ranked 70th

Apr 2026 - Present

- Estimated port position at approximately 5 mm MAE across five cases with YOLO keypoints and multiview triangulation. Reduced fine-grained XYZ/RPY MAE by 21.3%/16.5% versus the worst case using bilinear cross attention.
- Built LeRobot/YOLO collection with ROS timestamp checks; integrated ReID and post-occlusion target reacquisition.

CJ Logistics Challenge | Team Lead

Jul 2026 - Present

- Built pseudo-labeling, YOLOv8m-seg training and ONNX pipelines; tracked boxes and estimated W/D/H from monocular CCTV using a 62.3 cm rail-width reference.
- Configured an optional TensorRT FP16 execution path with engine and timing caches in ONNX Runtime.

RoboDine / Mobile Extension | Team project / Individual extension

Apr 2025 - Present

- Implemented YOLO/solvePnP object-pose estimation; computed Camera-LiDAR extrinsics from five-pose sensor recordings.
- Built person tracking and closest-approach-based slowing/stopping; evaluated IDF1, ID switches and sensor coverage in multi-person simulation.

Structural Stability | Dacon | Ranked 66th of 484 teams

Mar 2026

- Reduced validation log loss from 0.1216 to 0.0630 and raised accuracy from 96% to 99% with bidirectional cross attention versus feature fusion.
- Reduced log loss by 86.3% over baseline using VideoMAE features and distillation regularization.

Public-Sector Q&A Chatbot | VS Community | Individual project

Jan 2026 - Mar 2026

- Improved LLM-as-a-judge results by up to 22% and response speed by up to 66% using synthetic evaluation data, SFT, asynchronous vLLM inference and MCP.

EDUCATION

Sejong University

Mar 2019 - Feb 2026

Bachelor's Degree, Electronic Information and Communication Engineering

GPA: 4.02/4.5

- Major GPA: 4.13/4.5; graduated Feb 20, 2026.

AWARDS AND CREDENTIALS

Top Award, ROS2 & AI Autonomous Driving and Robotic Arm Bootcamp (Jun 2025)

Excellence Award, IN-JEJU Challenge Hackathon (Dec 2024)

Engineer Information Processing (Dec 2025); Class 2 Regular driver's license, South Korea (Nov 2020)

LANGUAGES

Korean: Native | **English:** OPIc IH (Mar 2026), TOEIC 925 (Jun 2025)